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Structure and stability of small copper clusters

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The structure and stability of small copper clusters with up to ten atoms has been determined both for the neutral and the ionic clusters with density functional calculations. The calculations were of all-electron type. The structure optimization and frequency analysis were performed on the local density approximation level with the exchange correlation functional by Vosko, Wilk, and Nusair. Subsequently improved calculations for the stability were based on the generalized gradient approximation, where the exchange correlation functional of Perdew and Wang was used. Finally, the binding energies, ionization potentials, electron affinities, and separation energies were calculated. The results show that the trends are in agreement with available experimental data. © *2002 American Institute of Physics.* [DOI: 10.1063/1.1436465]

I. INTRODUCTION

The study of clusters has become an increasingly interesting topic in recent years. This is due to their role as precursors of bulk material. In this sense the knowledge of their properties furnishes information on how the transition from atom or molecule to the solid state may occur. The determination of structural properties and the growth pattern of the clusters therefore offer a first step towards the understanding of the development of a solid. If we classify solids by their bonding properties, we can distinguish three categories with strong bonding, usually referred to as ionic, semiconducting or metallic solids. In this paper we want to address only the properties of metallic clusters. Here copper clusters offer a unique opportunity, because recent advances both in experimental and computational techniques can provide comparable information. Copper clusters with up to 5 atoms were studied a few years ago[1] with the density functional theory (DFT) program deMon.[2] In the meantime an efficient new DFT program ALLCHEM,[3] was developed which offered the opportunity to extend the size of clusters and to learn even more systematically how their growth can be understood. At the same time new experiments are now available[4–6] which allow a more quantitative comparison.

The goal of the present work is to devise a structural concept of copper clusters. In the following sections we present the structures of small copper clusters with up to 10 atoms. The stability of the structures is tested by a frequency analysis. Based on these results, structures of the corresponding cations and anions are also determined. Further analysis of the clusters furnishes information on their binding energies, ionization potentials, and electron affinities. Finally, the separation energies for the fragmentation of cationic clusters into smaller cationic clusters and neutral atoms or dimers are presented.

II. COMPUTATIONAL DETAILS

All presented calculations were carried out with the DFT program ALLCHEM, which is described in the habilitation thesis of one of the present authors.[7] The program uses the formalism of the Kohn–Sham equations[8] and analytic energy gradients for the structure optimization. It works with Gaussian-type basis sets and auxiliary functions, which allows one to avoid four-center integrals.[9] An efficient integral algorithm[10] was applied for the calculation of three-center Coulomb integrals, which exploits some special features of the used auxiliary functions.[11] In addition an adaptive grid[12] is used for the numerical calculation of the exchange-correlation potential. The structure optimizations were performed in the local density approximation (LDA) with the exchange-correlation functional of Vosko, Wilk, and Nusair (VWN)[13] and all-electron basis sets were employed. The same functional was used for the frequency analysis in order to be able to distinguish between stable structures and transition structures. LDA studies on small copper clusters with up to five atoms[1] have already shown that the DFT optimized double zeta plus valence polarization (DZVP) basis, together with the A2 auxiliary function set[14] supply very reliable geometries for structure optimizations. Since such calculations are insufficient for energetic considerations, single point energies were finally calculated using the same basis and auxiliary function set with the exchange-correlation functional of Perdew and Wang (PW).[15,16] This approach has already been proven reliable for small copper clusters with up to five atoms.

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TABLE I. Point group (PG), electronic state, structure parameters [Å] and frequencies [cm^{-1}] of neutral, cationic, and anionic copper clusters with up to five atoms. The cluster structures and atom numbers are given in Fig. 1.

Cluster	PG	State	Structure Parameters		Frequencies					
Cu_2	$D_{\infty h}$	$^1\Sigma_g$	$r_{12}=2.21$		269					
Cu_3	C_{2v}	$^2B_1$	$r_{12}=2.26$	$r_{23}=2.50$	118	177	290			
Cu_4	D_2	^1A_g	$r_{12}=2.36$	$r_{14}=2.24$	53	130	157	159	242	276
Cu_5	C_2	$^2A_1$	$r_{12}=2.32$	$r_{14}=2.36$	27	43	111	122	152	180
			$r_{24}=2.33$	$r_{45}=2.38$	222	235	277			
Cu_2^+	$D_{\infty h}$	$^2\Sigma_g$	$r_{12}=2.35$		215					
Cu_3^+	D_{3h}	$^1A_1'$	$r_{12}=2.30$		186	186	267			
Cu_4^+	D_{2h}	$^2B_{3u}$	$r_{12}=2.35$	$r_{14}=2.32$	8	141	142	172	232	254
Cu_5^+	D_{3h}	$^1A_1'$	$r_{12}=2.30$	$r_{14}=2.42$	104	104	116	116	166	195
					195	231	290			
Cu_2^-	$D_{\infty h}$	$^2\Sigma_u$	$r_{12}=2.34$		202					
Cu_3^-	$D_{\infty h}$	$^1\Sigma_g$	$r_{12}=2.30$		55	55	157	257		
Cu_4^-	D_{2h}	$^2B_{2u}$	$r_{12}=2.35$	$r_{14}=2.40$	50	128	133	180	231	231
Cu_5^-	C_{2v}	$^1A_1$	$r_{12}=2.30$	$r_{14}=2.39$	46	55	103	128	150	152
			$r_{24}=2.42$	$r_{45}=2.33$	206	232	275			

III. STRUCTURES AND FREQUENCIES OF COPPER CLUSTERS

A. Clusters with up to five atoms

The small copper clusters with up to five atoms were already extensively discussed.[1] Here they serve as test cases for the calculation of the larger clusters. Our optimized structure parameters and frequencies for the neutral and charged ground states of these clusters are listed in Table I. The corresponding cluster structures are depicted in Fig. 1 showing all bonds with bond lengths up to 2.6 Å. A more elaborate bond analysis will be presented in a separate paper.[17] The agreement between the previously calculated LDA structures[1] and the present ones is good. The largest deviation is 0.02 Å for r_{24} in Cu_5. If we compare our calculated Cu_2 bond length of 2.21 Å with the experimental value of 2.22 Å, we find it close and slightly better than the previous LDA value of 2.20 Å. Remarkable is the accuracy of the obtained frequency of 269 cm^{-1} (experiment: 265 cm^{-1}),[18] which is much better than the previous LDA value of 216 cm^{-1} and GGA value of 220 cm^{-1}. The deviations of the latter values can be ascribed to the numerical fitting of the exchange-correlation potential during the self-consistent field

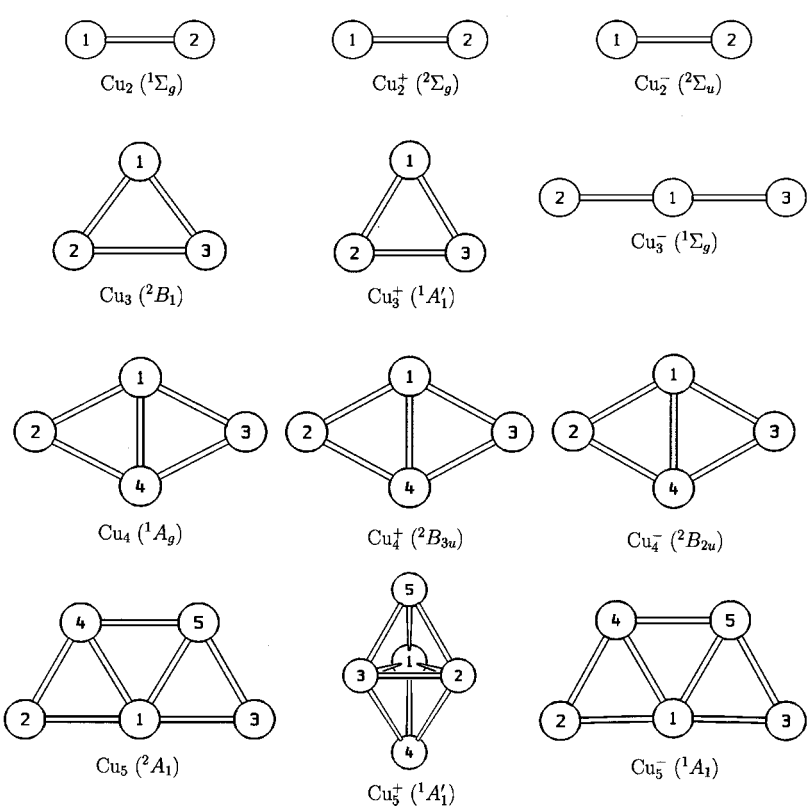

FIG. 1. Ground state structures of neutral, cationic and anionic copper clusters with up to five atoms.

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TABLE II. Point group (PG), relative LDA, and GGA energies [kcal/mol], structure parameters [Å], and frequencies [cm^{-1}] of neutral copper clusters with six to ten atoms. The cluster structures and atom numbers are given in Fig. 2.

Cluster	PG	LDA	GGA	Structure parameters		Frequencies					
Cu_6	D_{3h}	3.0	0.0	$r_{12}=2.33$	$r_{23}=2.39$	37	37	41	111	111	134
						134	183	211	228	277	279
Cu_6	C_{5v}	1.5	4.0	$r_{12}=2.41$	$r_{16}=2.32$	21	24	95	137	137	139
						139	179	179	240	261	263
Cu_6	C_{2v}	0.0	9.7	$r_{12}=2.36$	$r_{13}=2.47$	66	91	99	115	150	156
				$r_{15}=2.42$	$r_{56}=2.28$	165	173	182	225	257	283
Cu_7	D_{5h}	0.0	0.0	$r_{12}=2.39$	$r_{16}=2.39$	74	74	129	129	138	141
						141	161	184	184	189	189
						247	247	260			
Cu_7	C_{3v}	7.0	4.0	$r_{12}=2.41$	$r_{14}=2.42$	58	58	108	127	127	150
				$r_{15}=2.36$	$r_{45}=2.37$	150	152	170	191	191	197
						254	254	258			
Cu_8	C_{2v}	0.0	0.0	$r_{12}=2.35$	$r_{14}=2.47$	61	89	106	119	120	130
				$r_{15}=2.38$	$r_{23}=3.07$	131	132	135	151	196	197
				$r_{25}=2.39$	$r_{27}=2.38$	199	209	235	236	245	258
				$r_{57}=2.35$	$r_{78}=2.47$						
Cu_8	T_d	2.9	0.5	$r_{12}=2.36$	$r_{23}=2.43$	48	48	103	103	103	145
						145	145	164	167	167	167
						202	202	253	254	254	254
Cu_9	C_s	0.0	0.0	$r_{12}=2.39$	$r_{15}=2.42$	71	72	80	109	116	122
				$r_{16}=2.41$	$r_{17}=2.40$	125	127	132	138	140	147
				$r_{18}=2.42$	$r_{26}=2.46$	165	166	189	199	214	223
				$r_{27}=2.40$	$r_{28}=2.42$	250	253	283			
				$r_{45}=2.49$	$r_{46}=2.35$						
				$r_{47}=2.42$	$r_{68}=2.35$						
Cu_9	C_{2v}	3.1	1.1	$r_{12}=2.44$	$r_{15}=2.40$	56	64	67	93	97	119
				$r_{16}=2.41$	$r_{18}=2.37$	121	137	147	150	152	164
				$r_{34}=2.44$	$r_{36}=2.37$	166	177	184	201	215	249
				$r_{46}=2.43$	$r_{68}=2.38$	256	257	264			
Cu_9	C_{3v}	7.9	3.2	$r_{12}=2.38$	$r_{23}=2.36$	68	75	89	91	91	93
				$r_{27}=2.41$	$r_{78}=2.40$	106	109	131	131	161	167
						167	195	195	197	211	230
						259	274	274			
Cu_{10}	D_{4d}	0.0	0.0	$r_{12}=2.33$	$r_{15}=2.42$	33	33	99	101	101	102
				$r_{19}=2.45$		102	107	107	131	135	135
						160	188	188	210	210	219
						219	229	229	231	236	236
Cu_{10}	D_{2h}	9.7	8.2	$r_{12}=2.40$	$r_{15}=2.30$	50	58	71	80	91	95
				$r_{19}=3.50$	$r_{24}=2.50$	103	113	115	134	142	150
				$r_{28}=2.48$	$r_{35}=3.30$	151	172	175	179	185	186
						191	223	229	232	291	295

(SCF) procedure. The other calculated LDA frequencies agree better with the previous LDA values with differences not exceeding 20 cm^{-1}. This is within the accuracy of the method, since the second derivatives of the energy necessary for the frequency analysis are calculated numerically.

B. Neutral clusters

The optimized structure parameters, relative energies and vibrational frequencies of the low-lying minima of the larger neutral copper clusters are listed in Table II. The corresponding cluster structures are depicted in Fig. 2.

For the Cu_6 cluster we find a planar D_{3h} structure as global minimum. On the LDA level the three displayed Cu_6 structures are very close in energy and the three-dimensional C_{2v} structure is the global minimum. The GGA calculations separate these isomers energetically and shift the C_{2v} structure almost 10 kcal/mol above the planar D_{3h} ground state. This situation is similar to the neutral copper pentamer where the energy difference between the planar C_{2v} ground state (Fig. 1) and a three-dimensional D_{3h} structure is considerably increased by the gradient corrections. A similar conclusion was drawn from Car–Parinello studies of Cu_6[19] in which the same ground-state structure was obtained at the GGA level. Also in this study the energetical order of the isomers at GGA level changed compared to previous LDA calculations.[20] These calculations show that planar or nearly planar structures, like the depicted D_{3h} or C_{5v} structures, are favored for the neutral copper hexamer.

Different from the hexamer, the most stable neutral heptamer has a three-dimensional structure. We find in our calculations a pentagonal bipyramid of D_{5h} symmetry as global minimum for Cu_7 both on the LDA and GGA level. Compared with the C_{5v} structure of Cu_6, the ring distances of the Cu_7 cluster are slightly shortened from 2.41 Å to 2.39 Å and the cap-ring distance is substantially lengthened from 2.32 Å to 2.39 Å. Still the cap–cap distance is only 2.50 Å, indicat-

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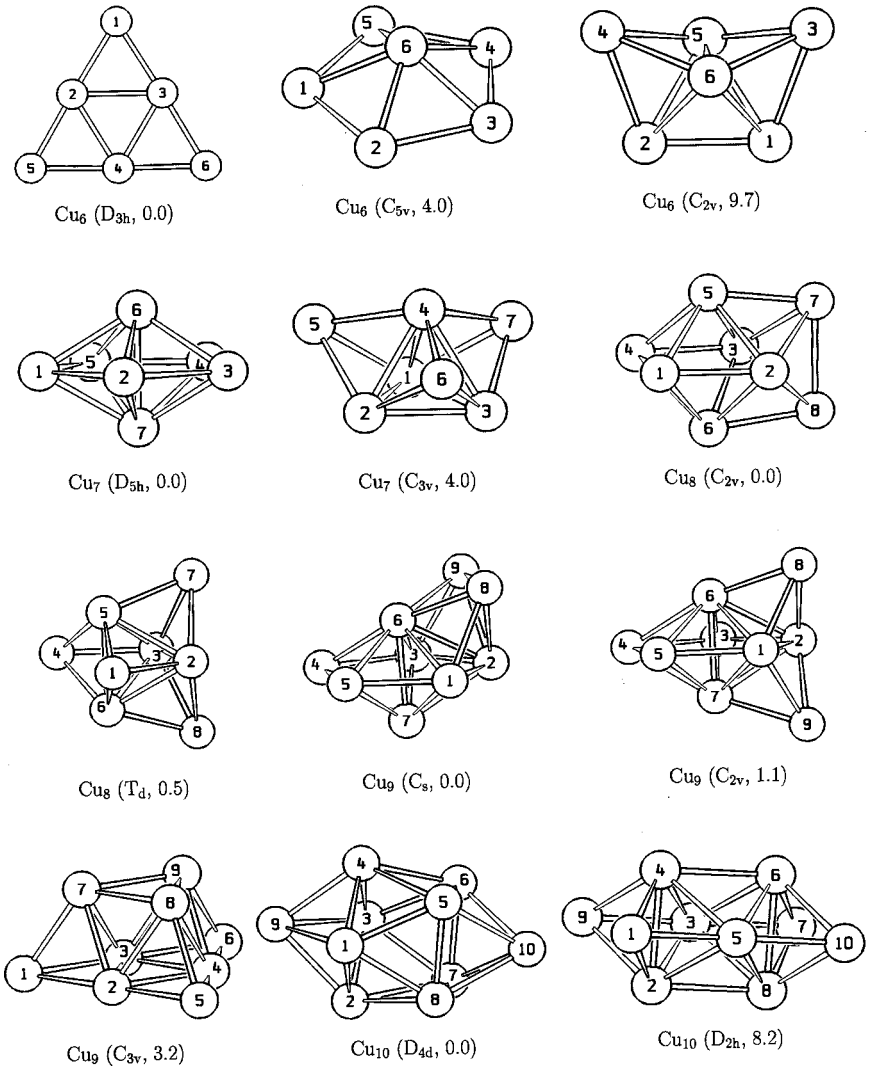

FIG. 2. Structures and relative GGA energies [kcal/mol] of neutral copper clusters with six to ten atoms.

ing a bond between the capping atoms. Another relatively stable Cu_7 isomer is the depicted C_{3v} structure in Fig. 2. This structure was obtained from the relaxation of a crystal fragment. During this relaxation, the average bond length decreased from 2.56 Å to 2.39 Å. Planar structures of the heptamer were found considerably higher in energy and can be excluded as possible ground states from our study.

To arrive at new cluster structures, the capping of stable structures seems to be a reasonable procedure. Starting from the Cu_7 bipyramid, a capping of one of the triangles has been proposed and calculated on the LDA level.[20] In our study we found that such a structure is not stable and relaxes. The global minimum of Cu_8 has C_{2v} symmetry (Fig. 2). This global minimum can be obtained from the relaxation of a capped triangle or a capped ring bond structure of the heptamer ground state. For Cu_8 a T_d structure is found to be only 0.5 kcal/mol higher than the C_{2v} structure. This structure has a cuboidal cage form compared with the more spherical cage form of the C_{2v} structure.

The search for stable structures of larger clusters is increasingly more difficult because of the increasing number of possible arrangements for isomers. Therefore, we have used the strategy of capping of smaller clusters or solid-state fragments. In the case of Cu_9 a starting structure for optimization can be obtained by single capping of Cu_8 structures or double capping of Cu_7 structures. The first procedure was accompanied by extremely strong relaxations, whereas the second procedure resulted in all cases in Cu_9 minima. The global minimum has C_s symmetry. It is derived from the Cu_7 bipyramid by double capping of two adjacent upper triangles. The five-membered ring and the bond between the capping atoms is preserved (Fig. 2). A C_{2v} structure with a slightly higher energy of 1.1 kcal/mol is obtained from double capping of adjacent upper and lower triangles. The C_{3v} isomer of Cu_9 depicted in Fig. 2 was obtained from the relaxation of a crystal fragment. The average bond distance of this isomer is similar to the C_{3v} isomer of Cu_7 and still considerably shorter than the bulk distance of 2.56 Å.

The concept of capping leads to stable structures also for Cu_{10} clusters. The global minimum has D_{4d} symmetry. Like the global minimum of Cu_8 this structure is a cage. The same structure was obtained in Ref. 20. It can be obtained by bicapping the T_d structure of Cu_8 over the two squares. A D_{2h} structure lies 8 kcal/mol higher in energy.

If we look at the trend for the bond distances of the ground-state structures with increasing cluster size, we find that the range of distances fluctuates. The differences be-

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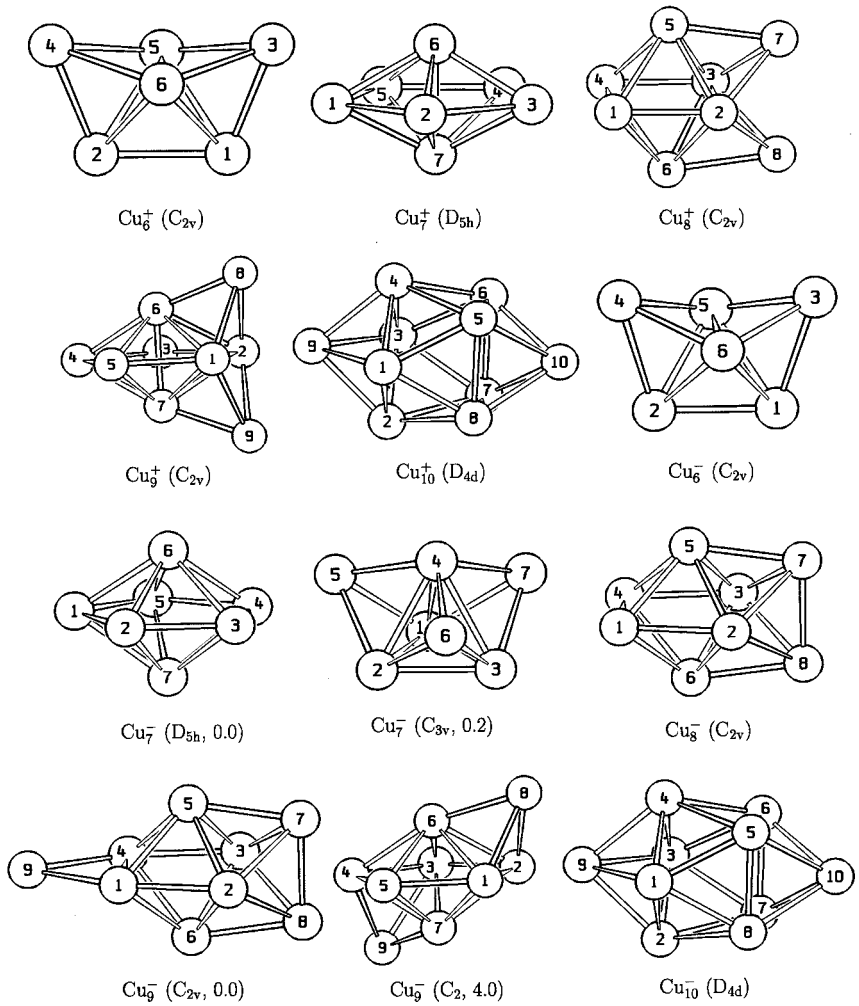

FIG. 3. Structures and relative GGA energies [kcal/mol] of charged copper clusters with six to ten atoms.

tween the largest and the smallest distances are 0.24 Å (Cu_3), 0.12 Å (Cu_4), 0.06 Å (Cu_5), 0.06 Å (Cu_6), 0.06 Å (Cu_7), 0.12 Å (Cu_8), 0.14 Å (Cu_9), and 0.12 Å (Cu_{10}). The frequencies range from 8 to 295 cm^{-1}. The maximum frequencies are within a range of 40 cm^{-1}.

C. Cationic clusters

The optimized structures of the global minima of the larger cationic Cu_n^+ clusters with $n=6-10$ are depicted in Fig. 3. Their optimized geometries and frequencies are listed in Table III. From Figs. 1, 2, and 3 we see that Cu_5, Cu_6 and Cu_9 undergo the most substantial structural changes upon ionization. The first two acquire a three-dimensional structure compared to the planar structure of the neutral molecules. In the case of Cu_9, the second lowest-lying neutral structure is found to be more favorable for the cation.

For Cu_2 there is a bond lengthening from 2.21 Å to 2.35 Å for the cation. The frequency is therefore reduced from 270 cm^{-1} to 215 cm^{-1}. This is in agreement with the experimental observation of a value of 187 cm^{-1}.[21] The bonding in the cation is reduced because a bonding orbital is only singly occupied. The Cu_3 undergoes a significant change in structure, too. The neutral molecule is an open-shell system which shows a Jahn–Teller distortion because of HOMO–LUMO degeneracy. The degeneracy is removed if the system is ionized, so that D_{3h} symmetry is possible. Since the average bond length is significantly reduced, the lowest frequency is increased from 118 to 186 cm^{-1} and shows a degeneracy. In contrast the increase of the bond length of the bridge bond in Cu_4^+ compared to Cu_4 causes a significant reduction of the lowest frequency which describes an out-of-plane motion. The structural changes in Cu_4, Cu_6, and Cu_9 upon ionization lead to frequency changes, too. The effect of ionization is less pronounced in the frequency spread for the clusters Cu_7, Cu_8, and Cu_{10}. However, it should be noted that the ionization of Cu_7 results in a shortening of the cap-cap distance from 2.50 Å to 2.31 Å and, consequently, in a flatter structure. In the case of Cu_8, ionization transforms the cage structure into a capped octahedral one. Thus, substantial structure relaxations accompany the ionization of Cu_7 and Cu_8.

D. Anionic clusters

Again the optimized structures of the global minima of anionic Cu_n^- clusters with $n=6-10$ are illustrated in Fig. 3. Their geometries and frequencies are listed in Table III. Compared to the neutral species, the most significant structural changes are observed for Cu_3^-, Cu_6^-, and Cu_9^-. The first system becomes linear, Cu_6^- acquires a three-

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TABLE III. Point group (PG), structure parameters [Å] and frequencies [cm^{-1}] of charged copper clusters with six to ten atoms. The cluster structures and atom numbers are given in Fig. 3.

Cluster	PG	Structure Parameters		Frequencies					
Cu_6^+	C_{2v}	$r_{12}=2.44$	$r_{13}=2.36$	81	85	104	131	147	150
		$r_{15}=2.42$	$r_{56}=2.28$	164	168	203	209	255	281
Cu_7^+	D_{5h}	$r_{12}=2.46$	$r_{16}=2.39$	89	89	114	114	140	140
				151	165	168	168	172	172
				250	250	279			
Cu_8^+	C_{2v}	$r_{12}=2.33$	$r_{14}=2.35$	48	81	85	102	104	118
		$r_{15}=2.47$	$r_{23}=2.49$	120	140	151	154	170	178
		$r_{25}=2.38$	$r_{27}=2.34$	194	208	240	249	254	271
		$r_{57}=2.47$	$r_{78}=2.77$						
Cu_9^+	C_{2v}	$r_{12}=2.40$	$r_{15}=2.43$	44	76	78	83	115	122
		$r_{16}=2.42$	$r_{18}=2.41$	130	140	141	145	149	150
		$r_{34}=2.45$	$r_{36}=2.40$	164	183	187	196	214	240
		$r_{46}=2.35$	$r_{68}=2.31$	254	271	292			
Cu_{10}^+	D_{4d}	$r_{12}=2.37$	$r_{15}=2.41$	47	47	96	96	98	98
		$r_{19}=2.43$		113	123	125	125	138	138
				144	185	185	202	202	212
				212	222	222	228	228	238
Cu_6^-	C_{2v}	$r_{12}=2.36$	$r_{13}=2.44$	52	85	119	134	139	155
		$r_{15}=2.40$	$r_{56}=2.48$	162	168	195	217	249	257
Cu_7^-	D_{5h}	$r_{12}=2.33$	$r_{16}=2.43$	41	41	116	120	120	137
				137	162	176	176	218	218
				235	235	245			
Cu_7^-	C_{3v}	$r_{12}=2.36$	$r_{14}=2.55$	35	45	107	128	128	136
		$r_{15}=2.38$	$r_{45}=2.32$	136	148	148	203	203	207
				246	246	250			
Cu_8^-	C_{2v}	$r_{12}=2.42$	$r_{14}=2.38$	56	89	94	101	117	120
		$r_{15}=2.41$	$r_{23}=3.04$	123	127	140	172	183	185
		$r_{25}=2.35$	$r_{27}=2.41$	202	210	225	226	241	246
		$r_{57}=2.42$	$r_{78}=2.38$						
Cu_9^-	C_{2v}	$r_{12}=2.39$	$r_{14}=2.39$	26	56	89	97	106	114
		$r_{15}=2.41$	$r_{19}=2.37$	123	124	129	132	148	156
		$r_{23}=3.04$	$r_{25}=2.36$	185	187	198	203	225	232
		$r_{27}=2.41$	$r_{57}=2.41$	238	243	252			
		$r_{59}=3.79$	$r_{78}=2.37$						
Cu_9^-	C_2	$r_{12}=2.37$	$r_{15}=2.38$	57	58	72	101	114	120
		$r_{16}=2.44$	$r_{17}=2.44$	121	124	136	141	143	156
		$r_{18}=2.40$	$r_{23}=2.39$	164	176	198	199	217	236
		$r_{26}=2.39$	$r_{27}=2.39$	246	251	252			
		$r_{28}=2.40$	$r_{56}=2.39$						
		$r_{57}=2.39$	$r_{68}=2.42$						
Cu_{10}^-	D_{4d}	$r_{12}=2.35$	$r_{15}=2.43$	51	51	96	96	97	100
		$r_{19}=2.44$		100	119	119	127	132	132
				156	190	190	207	207	212
				212	219	219	226	231	231

dimensional structure like the corresponding cation and the Cu_9^- ground state represents a bond ring capping of Cu_8 which is not stable for the neutral nonamer.

The frequency reduction for Cu_2^- to 201 cm^{-1} compares well with experiment[22] where a value of 210 ± 15 cm^{-1} is found. It is due to a bond lengthening which has its origin in the occupancy of an antibonding orbital. The Cu_3^- cluster is linear with a bond length close to the one in Cu_3^+. However, since the latter structure is an equilateral triangle, its frequencies are much larger. In Cu_4^- the bridge bond is even more lengthened than in Cu_4^+ compared to Cu_4. The changes from Cu_5 to Cu_5^- are minor, both in geometry and frequency.

Although the Cu_6^- structure looks like the Cu_6^+ structure, their corresponding bond distances are quite different and so are their frequencies. The Cu_7^- ground state possesses D_{5h} symmetry and is topologically similar to the corresponding neutral cluster. The main difference is the cap-cap distance which is considerably larger in the anion with a value of 2.80 Å. As a consequence the bond between the cap atoms disappears and a cage structure is obtained. Only 0.2 kcal/mol above this ground state there is a C_{3v} structure of the heptamer anion depicted in Fig. 3, too. It is similar to the corresponding neutral structure with an increased r_{14} distance. The near degeneracy of these two structures was also found in a Car–Parinello study.[23] For the Cu_9^- we find a C_{2v} structure as global minimum, which has no topological equivalent for the neutral cluster. This structure is 4 kcal/mol more stable than the C_2 structure also depicted in Fig. 3. The latter was assigned as the Cu_9^- ground state in a Car–Parinello study.[23] The cluster structures of Cu_8^- and Cu_{10}^- are similar to their corresponding neutral clusters and retain their symmetry.

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TABLE IV. Binding energies (BE), basis set superposition error (BSSE), zero point energies (ZPE), and corrected binding energies (CBE) per atom [kcal/mol] of neutral copper clusters.

Cluster	BE/Atom	BSSE/Atom	ZPE/Atom	CBE/Atom	Expt.		
Cu_2	25.6	1.4	0.2	24.0	24.0[a]	23.5±1.96[b]	20.8 ± 1.68[a]
Cu_3	25.0	2.1	0.3	22.6	23.4[d]	24.5±2.86[b]	19.7 ± 2.90[c]
Cu_4	35.5	2.9	0.4	32.2		34.0±3.29[b]	26.5 ± 6.53[c]
Cu_5	38.3	3.5	0.4	34.4		35.8±3.40[b]	28.5 ± 8.86[c]
Cu_6	43.0	4.1	0.4	38.5		39.6±4.14[b]	30.7 ± 10.30[c]
Cu_7	45.0	4.9	0.5	39.6		42.7±5.00[b]	29.8 ± 11.41[c]
Cu_8	47.6	5.3	0.5	41.8		46.0±5.34[b]	32.3 ± 12.26[c]
Cu_9	47.6	5.7	0.5	41.4			31.3 ± 8.73[c]
Cu_{10}	48.9	5.9	0.5	42.5			

[a]From Ref. 26.
[b]From Ref. 4.
[c]From Ref. 5.
[d]From Ref. 27.

IV. CLUSTER ENERGETICS

A. Binding energies

For the optimized structures of the neutral and ionic clusters, binding energies per atom were calculated at the GGA level. Recently, it was pointed out that basis set superposition errors may be important for a quantitative calculation of cluster binding energies.[24] Therefore, we have calculated the basis set superposition error (BSSE) per atom using the approximated correction by Jansen and Ross.[25] It was shown[24] that the accuracy of this approximation is within 0.2 kcal/mol per atom for the used basis set. We also have calculated the zero-point energies (ZPE) of the clusters. The values for the neutral clusters are listed in Table IV. As can be seen from this table the BSSE per atom increases from 1.4 kcal/mol for Cu_2 to 5.9 kcal/mol for Cu_{10}. The ZPE per atom is much less size dependent and considerably smaller. For Cu_2 and Cu_3, direct measurements of the atomization energies by fluorescence[26] and Knudsen cell mass spectrometry[27] are available. Table IV shows that our corrected binding energies (CBE) per atom are in good agreement with these direct measurements.

Very recently the binding energies of the larger neutral clusters were derived from collision-induced dissociation (CID) experiments of anionic[4] and cationic[5] copper clusters using electron affinities or threshold detachment energies[28] and ionization potentials[29–32] from the literature. In Fig. 4 the derived binding energies from the CID of anionic (■) and cationic (▲) copper clusters are compared with our calculated values (●). This figure shows that our corrected binding energies are in good agreement with those derived from the CID of anionic copper clusters. All theoretical values are within the error bounds of the experiment.

Also, the binding energies derived from the CID of the cationic clusters show the same trend as the calculated values. However, they are considerably smaller. Due to the relatively large uncertainty in the ionization potentials, the theoretical values are still within the error bounds. As a result of the BSSE correction, the size dependence of the difference between the calculated and experimental binding energies as discussed in recent work[5] has been reduced here.

The binding energies per atom show an almost monotonic increase with cluster size. Exceptions are the Cu_2 and Cu_8 which possess higher binding energies per atom than Cu_3 and Cu_9, respectively.

The binding energies per atom of the cationic copper clusters are listed in Table V and depicted in Fig. 5. For Cu_2^+ and Cu_3^+ they can be derived from the literature values of the neutral dimer,[26] and trimer[27] in combination with the well-established values for the ionization potentials of the atom, dimer, and trimer. Our corrected binding energies are about 3 kcal/mol higher than these derived values. The experimental CID values (▲ in Fig. 5) from Ref. 5 are considerably smaller. For Cu_4^+ the discrepancy between the calculated and measured binding energies per atom increases to almost 10 kcal/mol. This difference slightly increases until 14 kcal/mol for Cu_9^+. Thus, we can conclude that the relatively small binding energies per atom derived for the neutral clusters from the CID measurements of the cations have their origin in the smaller experimental values for the cations compared to theory.

As Fig. 5 shows, the agreement between theory and experiment for the general trend of the binding energy per atom

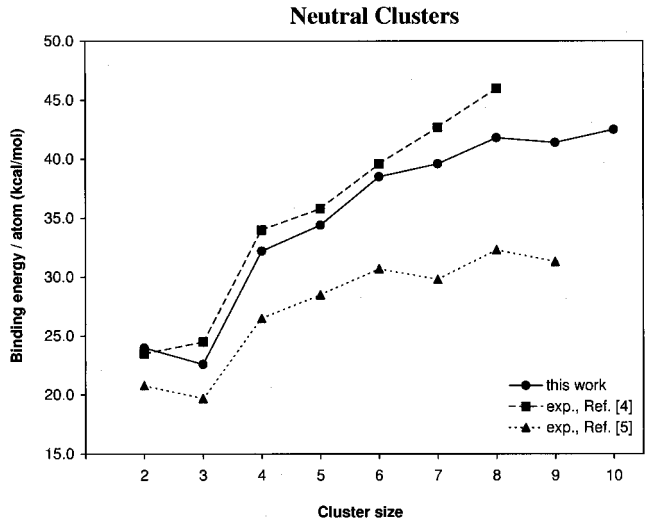

FIG. 4. Binding energies per atom [kcal/mol] for neutral copper clusters.

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TABLE V. Binding energies (BE), basis set superposition error (BSSE), zero point energies (ZPE), and corrected binding energies (CBE) per atom [kcal/mol] of cationic copper clusters.

Cluster	BE/Atom	BSSE/Atom	ZPE/Atom	CBE/Atom	Expt.	
Cu_2^+	26.2	1.4	0.2	24.6	21.9[a]	18.9±1.60[b]
Cu_3^+	43.6	2.2	0.3	41.1	38.2[a]	34.5±2.78[b]
Cu_4^+	42.9	3.0	0.4	39.5		29.8±2.77[b]
Cu_5^+	46.3	3.9	0.4	42.0		35.1±4.06[b]
Cu_6^+	48.4	4.4	0.5	43.5		32.9±3.19[b]
Cu_7^+	52.2	5.0	0.5	46.7		35.1±3.57[b]
Cu_8^+	52.4	5.5	0.5	46.4		33.9±3.86[b]
Cu_9^+	54.7	5.8	0.5	48.4		34.4±4.14[b]
Cu_{10}^+	53.9	6.0	0.5	47.4		

[a]Derived from literature values (see text).
[b]From Ref. 5.

is good. We find pronounced stability for Cu_3^+, Cu_7^+, and Cu_9^+. If only the trend, and not the absolute values, is considered, we see a similar trend of the binding energies for the Cu_n and the Cu_{n+1}^+ clusters. This remarkable similarity can be explained by the following observation. If the full $3d$ shell is ignored and only the s valence electrons are counted, Cu_n and Cu_{n+1}^+ can be considered as isoelectronic. This is reflected in the binding energies.

The binding energy per atom of the anionic copper clusters is listed in Table VI and compared with the experimental CID values in Fig. 6. The agreement is quite satisfactory. All theoretical values are within the experimental uncertainties. This is also the reason for the good agreement between the theoretical and experimental binding energies of neutral clusters derived from the CID measurements of the anions (■ in Fig. 4). Because this derivation includes the approximation of the electron affinities by threshold detachment energies, we can conclude that this approximation is reliable. The obtained binding energies agree well with our calculated values and possess relative small uncertainties.

For the neutral clusters we are inside the error ranges of both the experimental values reported in Refs. 4 and 5, respectively, with the exception of Cu_2 in Ref. 5.

For the cationic clusters our values are considerably larger than the reported experimental data which is in accordance with the discussed trend of the cationic CID derived bond energies of neutral clusters. The direct comparison of the bond energies of the cationic clusters suggests that the dissociation threshold from Ref. 5 is too small from Cu_6^+ on. This is in agreement with the cationic CID results reported by Krückeberg *et al.*

With the anionic clusters we are in the error range of the experiment. We can conclude that the agreement with the anionic CID values is good.

B. Ionization potentials and electron affinities

Adiabatic and vertical ionization potentials (IP) obtained from LDA calculations are presented in Table VII and compared with the available experimental data.[29–33] The latter are given with their error bounds. Here we present LDA IPs in order to compare directly with the values in Ref. 1. The inclusion of gradient corrections does not improve the calculated value and results in the same trend. As already discussed in previous work,[1] the calculated IPs are too large.

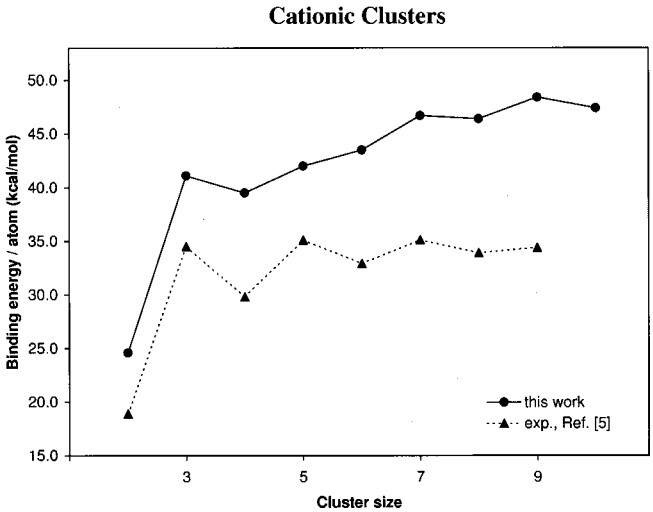

FIG. 5. Binding energies per atom [kcal/mol] for cationic copper clusters.

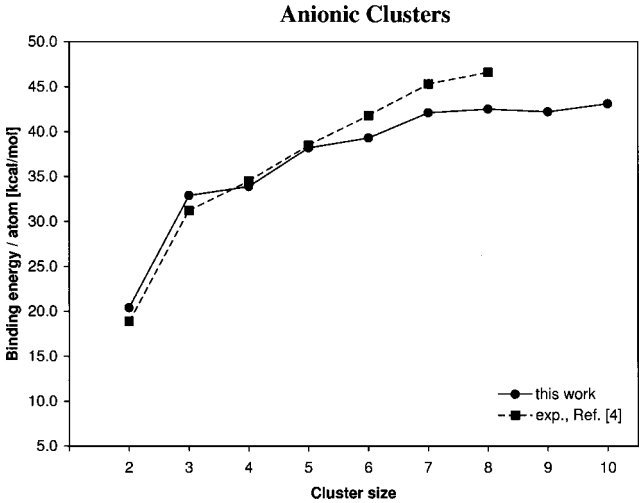

FIG. 6. Binding energies per atom [kcal/mol] for anionic copper clusters.

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TABLE VI. Binding energies (BE), basis set superposition error (BSSE), zero point energies (ZPE), and corrected binding energies (CBE) per atom [kcal/mol] of anionic copper clusters.

Cluster	BE/Atom	BSSE/Atom	ZPE/Atom	CBE/Atom	Expt.[a]
Cu_2^-	21.7	1.1	0.2	20.4	18.9±1.73
Cu_3^-	35.0	1.9	0.2	32.9	31.2±2.31
Cu_4^-	37.2	2.9	0.4	33.9	34.5±2.42
Cu_5^-	42.1	3.5	0.4	38.2	38.5±2.49
Cu_6^-	44.1	4.3	0.5	39.3	41.8±3.23
Cu_7^-	47.4	4.8	0.5	42.1	45.3±3.49
Cu_8^-	48.2	5.2	0.5	42.5	46.6±3.92
Cu_9^-	48.2	5.5	0.5	42.2	
Cu_{10}^-	49.5	5.9	0.5	43.1	

[a]From Ref. 4.

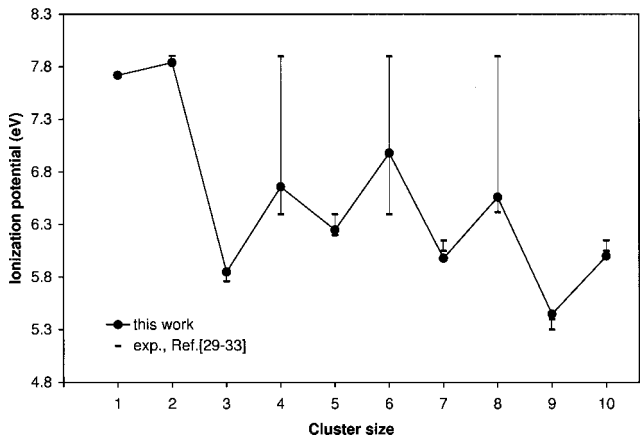

FIG. 7. Ionization potentials [eV] of copper clusters.

Therefore, we have scaled these values with the ratio between the atomic experimental IP and the calculated LDA value (0.898).

As Table VII shows our calculated IPs agree well with the previous LDA values[1] for the small clusters. The scaled vertical IPs are for all clusters in reasonable agreement with experiment. In Fig. 7 we compare our scaled vertical IPs with the experimental values. As can be seen, the scaled IPs are reliable and may be used in order to obtain binding energies of neutral clusters from the CID of cations. This would reduce the uncertainty of this approach for the neutral binding energies considerably.

The overall trend of the ionization potentials shows a remarkable oscillation. The values for clusters with an odd number of atoms are decreased compared with the even-numbered clusters. This leads to maxima for even-numbered and minima for odd-numbered clusters. This oscillation has been called even-odd effect in the literature.[34]

The corresponding plot for electron affinities (EA) is shown in Fig. 8. The calculated adiabatic and vertical EAs are compared with the available experimental data[22,28,35–37] in Table VIII. The scaled values are obtained analogous to the scaled IPs using a factor of 0.831. As Fig. 8 shows, the scaled vertical EAs are in good agreement with the experimental values which are vertical detachment energies for the larger clusters. Exceptions are Cu_9 and Cu_{10}, where the theoretical and experimental values differ by more than 0.5 eV. We cannot exclude that lower minima exist on the anionic potential-energy surface for these systems.

Similar to the ionization potentials, the electron affinities show a characteristic oscillation. However, the electron affinities show minima for the even-numbered clusters and maxima for the odd-numbered clusters.

The following explanation for the oscillatory trends can be given. Even-numbered copper clusters are closed-shell systems, whereas odd-numbered copper clusters are open-shell systems. For the latter systems, the single electron in the highest occupied molecular orbital (HOMO) can be more easily removed than an electron from the doubly occupied HOMO of a closed-shell system. In consequence, the ionization potentials in odd-numbered systems show a minimum and in even-numbered systems a maximum. Just the opposite is true for electron affinities. The clusters can more easily acquire an electron in the open-shell HOMO of the odd-numbered system than in the LUMO of a closed-shell system. This leads to maxima for odd-numbered systems and minima for even-numbered systems.

TABLE VII. Adiabatic and vertical ionization potential (IP) of the calculated copper cluster ground states at LDA level. All values are in eV.

	Adiabatic		Vertical		
Cluster	IP	Scaled IP	IP	Scaled IP	Expt.
Cu	8.60	7.72	8.60	7.72	7.724[a]
Cu_2	8.67	7.78	8.73	7.84	7.9042±0.0006[b]
Cu_3	6.44	5.78	6.52	5.85	5.80±0.04[c]
Cu_4	7.40	6.64	7.42	6.66	7.15±0.75[d]
Cu_5	6.58	5.91	6.96	6.25	6.3±0.1[c]
Cu_6	6.91	6.20	7.77	6.98	7.15±0.75[d]
Cu_7	6.56	5.89	6.66	5.98	6.1±0.05[c]
Cu_8	7.11	6.38	7.31	6.56	6.42 to 7.90[e]
Cu_9	5.97	5.36	6.07	5.45	5.35±0.05[c]
Cu_{10}	6.63	5.95	6.68	6.00	6.1±0.05[c]

[a]From Ref. 29.
[b]From Ref. 30.
[c]From Ref. 31.
[d]From Ref. 32.
[e]From Ref. 33.

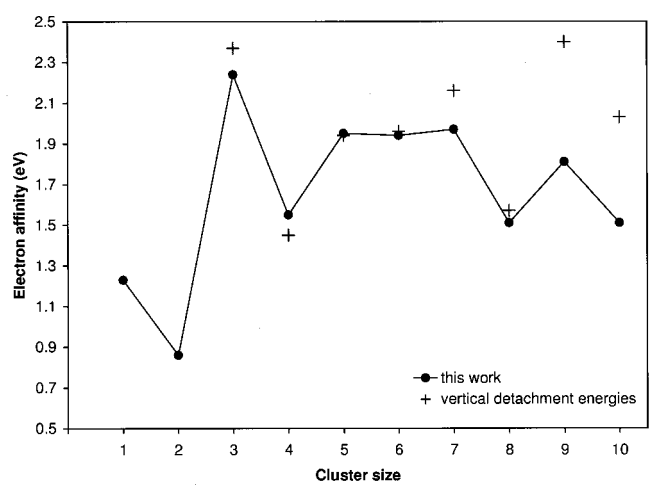

FIG. 8. Electron affinities [eV] of copper clusters.

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TABLE VIII. Adiabatic and vertical electron affinities (EA) of the calculated copper cluster anions at LDA level. All values are in eV.

	Adiabatic		Vertical		
Cluster	EA	Scaled EA	EA	Scaled EA	Expt.[a]
Cu^-	1.48	1.23	1.48	1.23	1.235 ± 0.005[b]
Cu_2^-	1.09	0.91	1.15	0.96	0.836 ± 0.006[b]
Cu_3^-	2.29	1.91	2.69	2.24	2.37 ± 0.01[c]
Cu_4^-	1.79	1.49	1.86	1.55	1.45 ± 0.05[c]
Cu_5^-	2.30	1.91	2.34	1.95	1.94 ± 0.05[c]
Cu_6^-	2.24	1.86	2.33	1.94	1.96 ± 0.05[c]
Cu_7^-	2.25	1.87	2.37	1.97	2.16 ± 0.1[c]
Cu_8^-	1.73	1.44	1.82	1.51	1.57 ± 0.05[c]
Cu_9^-	1.58	1.32	2.17	1.81	2.40 ± 0.05[c]
Cu_{10}^-	1.80	1.50	1.82	1.51	2.03 ± 0.05[c]

[a]From Ref. 28.
[b]Electron affinity.
[c]Vertical detachment energy.

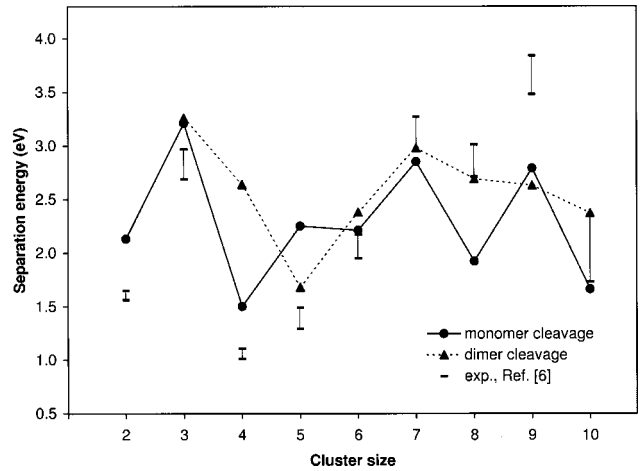

FIG. 9. Separation energies [eV] of copper clusters for monomer and dimer cleavage.

C. Separation energies and fragmentation channels for cationic clusters

Separation energies are defined as energies which are necessary to cleave a cluster in two or more fragments. The simplest fragmentations would be the cleavage of a Cu atom or a Cu_2 dimer. This has been observed in experiments. Photofragmentation experiments yielded fragmentation pathways of Cu_n^+ for $n=2-8$.[38,39] Collision induced dissociation experiments of small copper cluster anions and cations were recently presented.[4–6] Decay pathways and dissociation energies are presented,[6] for copper cluster cations over a large range, namely $n=2-25$. In this experiment positively charged clusters are prepared by laser vaporization and trapped in a magnetic field. These clusters decay by collision with argon atoms under cleavage of Cu or Cu_2. In this section we will compare our predicted fragmentation pathways with those deduced from experiment. For this purpose we have used the following equations for the determination of the separation energy for monomer and dimer cleavage

$$D(n^+,1^0)=E_B(n^+)-E_B([n-1]^+),$$
$$D(n^+,2^0)=E_B(n^+)-E_B([n-2]^+)-E_B(2). \quad (1)$$

Here $D(n^+,m^0)$ is the separation energy for a cationic cluster with n atoms under cleavage of a monomer ($m_0=1$) or dimer ($m_0=2$). The needed separation energy is then obtained as the difference of the binding energies of the fragments involved. In the case of dimer cleavage, the binding energy $E_B(2)$ of Cu_2 has to be included. Figure 9 shows the separation energies for monomer (●) and dimer (▲) cleavage in dependence of the cluster size. The experimental values[6] are presented with their error bounds. The decay channel in the experiment is derived from the intensities of the mass spectra.

From the calculations we can derive a preference for monomer cleavage, if the separation energy for this process is lower than for the dimer cleavage. This is the case for the even-numbered cations both in the calculation and the experiment. For the odd-numbered cations, the situation is more complicated. The experiments predict preference for dimer cleavage for Cu_3^+ and Cu_5^+, both monomer and dimer cleavage for Cu_7^+, and monomer cleavage for Cu_9^+. The calculations arrive at the same answer only for Cu_5^+. For Cu_3^+ the energy difference between $D(n^+,1^0)$ and $D(n^+,2^0)$ is only 0.05 eV and, thus, both monomer and dimer cleavage is predicted. In the case of Cu_9^+, we find from our calculations a preference for dimer cleavage, whereas monomer cleavage is found for Cu_7^+. In the latter case it is not clear from the experiments whether the dimer cleavage is a competing process or a consecutive reaction.[40]

Discrepancies between experimental and calculated reaction channels can be due to the fact that the experiments deal with highly excited clusters, whereas the calculations refer to ground states. For highly excited clusters, consecutive reactions cannot be excluded.

V. CONCLUSION

DFT calculations on the LDA level have proved to be a suitable means for structure and frequency determination of small copper clusters. GGA corrections are needed for the correct prediction of the relative stability of isomers and the global minimum. Similar trends are found for the binding energies of Cu_n and Cu_{n+1}^+ clusters, which can be explained by the isoelectronic nature of the valence *s* orbitals. It was shown that the BSSE increases with the cluster size. Thus, BSSE corrections are crucial for the quantitative comparison with experiment. For the anionic copper cluster, we find our corrected binding energies in good quantitative agreement with the recent experimental data.[4] The same holds for the binding energies of the neutral clusters derived from these experiments.

The ionization potentials and electron affinities show oscillatory behavior for even- and odd-numbered clusters because of their closed- and open-shell HOMOs in agreement with experiment. Finally, separation energies allow to predict fragmentation channels for monomer and dimer cleavage. Here the agreement with experiment is less satisfactory.

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